Danesh Morales Hashemi

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Research Positions

2025 – Present — **Graduate Research Assistant**, Institute for Quantum Computing
Advisors: Graeme Smith and Ningping Cao

2024 – 2025 — **Undergraduate Research Assistant**, Institute for Quantum Computing
Advisor: Joseph Emerson
Topic: *Circuit Benchmarking for Efficient and Scalable Fidelity Estimation*

Fall 2023 — **Undergraduate Research Assistant**, University of Waterloo
Advisor: Roberto Guglielmi
Topic: *Controllability of Fokker-Planck Equation*

Education

2025 – Present — **MMath in Applied Mathematics**, University of Waterloo
Quantum Information Specialization

2020 – 2025 — **BMath in Mathematical Physics**, University of Waterloo
Co-op Program • Pure Mathematics Minor • Graduated with Dean's Honours

Publications

1. **What Must a Quantum-Memory Decoder Know About Temporally Correlated Noise?**
   Danesh Morales-Hashemi, Graeme Smith, and Ningping Cao.
   Preprint, 2026 • arXiv:2610.03545 [quant-ph]

Conferences and Talks

1. **Demonstrating Circuit Benchmarking for Scalable and Efficient Fidelity Estimation**
   APS March Meeting 2025
   Session on Scalable Verification and Validation (QCVV)
   Anaheim Convention Center • March 2025

Teaching Experience

Spring 2025	**Teaching Assistant**, AMATH 250 – Introduction to Differential Equations
Fall 2023	**Instructional Support Assistant**, MATH 115 – Linear Algebra for Engineering
Winter 2022	**Instructional Support Assistant**, CS 115 – Introduction to Computer Science

Projects

1. **Introduction to Cycle Benchmarking for Quantum Error Diagnostics**
   QIC 820 Final Project, Institute for Quantum Computing • December 2025
2. **Predicting MLB Playoffs Using Quantum Machine Learning**
   QIC 890 Final Project, Institute for Quantum Computing • December 2025
   GitHub repository
3. **Demonstrating Circuit Benchmarking for Efficient and Scalable Fidelity Estimation**
   Undergraduate Thesis, University of Waterloo • April 2025
4. **Introduction to Superoperator Representations and Characterization of Completely Positive Maps**
   AMATH 900 Final Project, University of Waterloo • August 2024

Awards and Scholarships

Institute for Quantum Computing Undergraduate Research Award 2024	**$16,000 CAD**
University of Waterloo President's Scholarship 2020	**$2,500 CAD**
Full Undergraduate Educational Excellence Scholarship 2019, SENACYT, Panama	**$250,000 USD**

Relevant Coursework

(with links to syllabus)

Quantum Physics	Quantum Theory 1/2/3	Representation Theory
Quantum Information Processing	Quantum Machine Learning	Groups and Ring Theory
Quantum Devices	Theory of Quantum Information	Galois and Field Theory
Neural Networks	Functional Analysis	Differential Geometry